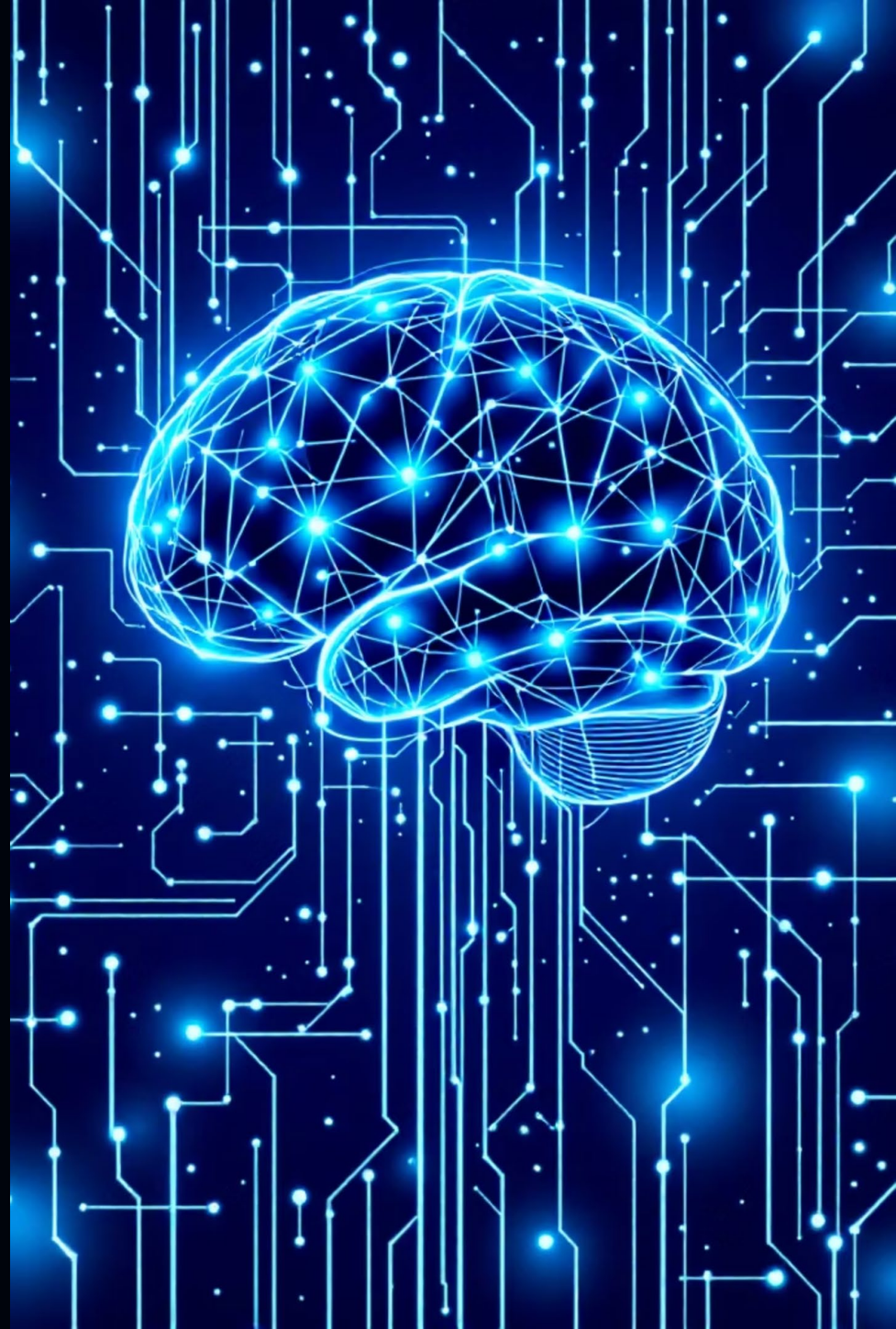
Real-Time Deep Learning Classifier for Multiclass Motor Imagery BCIs

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Background and Significance

Motor imagery Brain-Computer Interfaces show significant promise in neurorehabilitation and assistive technology. However, a critical challenge stands still: achieving high decoding accuracy in multiclass scenarios.

While binary classifications (e.g., left vs. right hand movements) show success, transitioning to complex functional tasks remains highly difficult. This research bridges the gap between theoretical deep learning capacity and practical, high-accuracy BCI control.





Project Goals

Primary Objective

Develop a robust deep learning classifier for real-time decoding of nine distinct hand movements using EEG signals.

Classification Model

Establish high-accuracy multiclass classification by overcoming overfitting issues in standard neural network architectures.

Signal Analysis

Analyze relationships between non-invasive EEG and invasive ECoG to inform feature extraction strategies.

Proposed Plan

Implementing a comprehensive neural network pipeline within a Python-based environment using BIDS standard for reproducibility.

01

Environment Setup

Fork and clone repository, create isolated Conda environment for dependency management.

02

Data Organization

Organize datasets using BIDS standard for consistent formatting and reproducibility.

03

Experimental Testing

Test full 9-class multiclass dataset and simplified binary subset (Finger vs. Wrist movements).

04

Model Development

Write custom classifier with advanced preprocessing and regularization techniques to enhance generalization.



Expected Outcomes and Deliverables

Optimized Decoder

Fully optimized multiclass model ready for real-time application with high classification accuracy.

Overcoming Baseline Issues

Resolved overfitting issues and improved multiclass separation capabilities with limited data.

Robust Method

Reliable deep learning method that translates complex motor imagery into actionable BCI commands.